

Work Hardening (strain hardening)

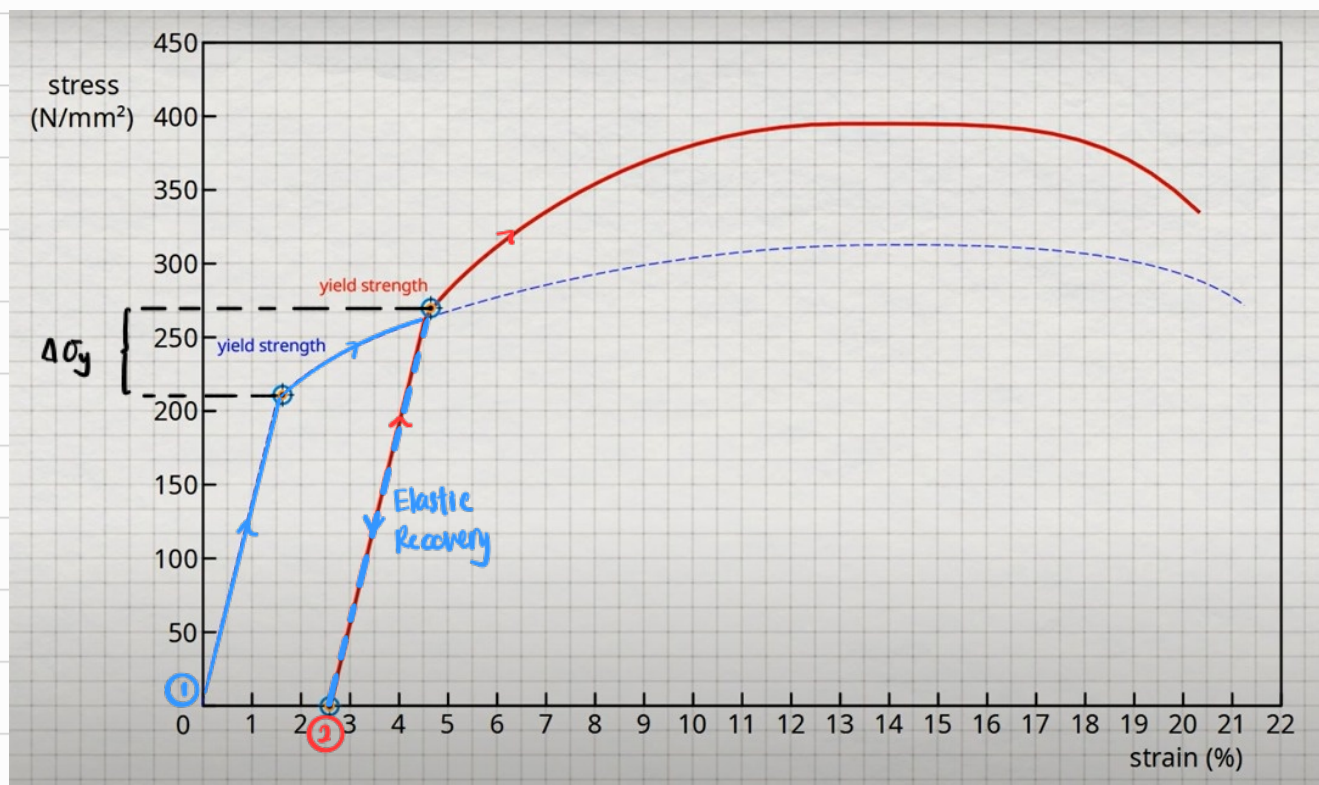
A metal gets plastically deformed when layers of atom slip past each other by breaking & forming bonds causing dislocations

However the more dislocations, the harder for layers of atom to slip, thus increasing the yield strength.

∴ We can intentionally introduce dislocations by plastically deforming the metal to increase the yield strength [strain hardening]

[However at the maximum dislocation density, if more dislocations are added, the material will break]

Example : Bending a paper clip front & back until it breaks.



Work Hardening

<https://youtu.be/54H5HQIMEio?si=0pOzDWtQflyyWj-y>